

Title: “Neural Mass Dynamics: From Critical Phase Transitions to Reaction-Diffusion Parameter Modulation in Alzheimer's Tau Propagation”

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Abstract: There is currently no mathematical basis for why Alzheimer's disease (AD) tau pathology first targets certain brain regions such as the transentorhinal and entorhinal cortex. Hubness is the leading structural explanation for this phenomenon. However, hub degree is confounded with a region's dynamical centrality, so connectivity alone cannot show whether a region's dynamics drives pathology, as opposed to its wiring. Here we rank brain regions by their dynamical resistance to destabilization, linearizing a Wilson-Cowan neural mass model near criticality on an 83-region HCP connectome and applying network control theory to compute each region's minimum control energy. Across 685 ADNI subjects, tau accumulated preferentially in these hardest-to-tip regions ($\rho \approx -0.29$ to -0.31), the opposite of amyloid deposition ($\rho \approx +0.29$), and the relationship strengthened after controlling for hub degree ($\rho \approx -0.40$). We then use this dynamical ranking to inform a personalized Fisher-Kolmogorov reaction-diffusion model, calibrated to 43 subjects' longitudinal tau-PET trajectories, and simulate five region-targeted protection strategies. Reducing local tau growth outperformed reducing global tau diffusion, and protecting the regions with the highest baseline tau burden produced the largest reduction in simulated tau burden (17.7–20.5%), exceeding protection of the dynamically hardest-to-tip regions (3.9–7.0%). Our framework identifies where tau pathology emerges and shows that burden-informed region selection outperforms structural or dynamical criteria for guiding future anti-tau interventions.